

Intro to Scratch

PLAY



What is Scratch?

Scratch is a programming language with a simple visual interface that allows for creation upon digital stories, games and animations. It is a universally used program commonly used among those learning code for the first time.

The scratch mascot:
Scratch Cat



Signing Up

It is very simple to make your own account!

In your web browser:

Look up Scratch and click on the first link.

When on the Scratch website, click join Scratch!

You will need:

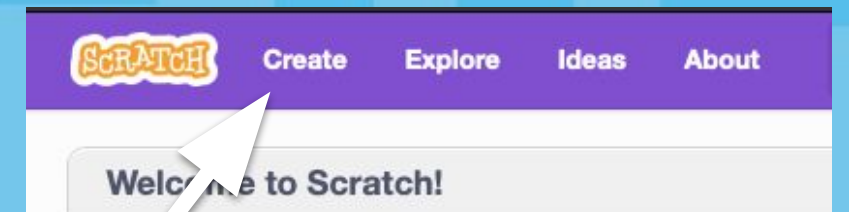
A username

A password

and an email

Let's do that now!

Let's create our first Scratch program



Let's hit the Create button

Scratch

Settings

File

Edit

Untitled-2

See Project Page

Tutorials

Gingy68

Code

Costumes

Sounds

Motion

Looks

Sound

Events

Control

Sensing

Operators

Variables

My Blocks

Motion

move 10 steps

turn 15 degrees

turn 15 degrees

go to random position

go to x: 0 y: 0

glide 1 secs to random position

glide 1 secs to x: 0 y: 0

point in direction 90

point towards mouse-pointer

change x by 10

set x to 0

change y by 10

set y to 0

if on edge, bounce

set rotation style left-right

x position

Green flag and a red stop sign

Notice...

This is what your Scratch project will look like when you hit the create button

What might we see?

A bunch of code block snippets

A big white box

What does this all mean? What is it?

Green flag and a red stop sign

Scratch Cat

A white box with Scratch Cat

Sprite1

x 0 y 0

Show

Size 100

Direction 90

Sprite1

Stage

Backdrops

1

Backpack

Dive Deeper

The Canvas

You can zoom in, zoom out and you can put any code blocks in this area.

Click and drag blocks in and to get rid of them on the canvas, you can drag back where you got it from or right click and delete it.

The Green and Red Stop

The green flag in scratch works as a run button. When you have code blocks down by hitting this it will run everything at once.

The red stop sign works like a stopper. This makes all of the code stop.

The Code Blocks

In the code tab we have quite a few tools at hand.

Motion

Looks

Sound

Events

Control

Sensing

Operators

Variables

My Blocks

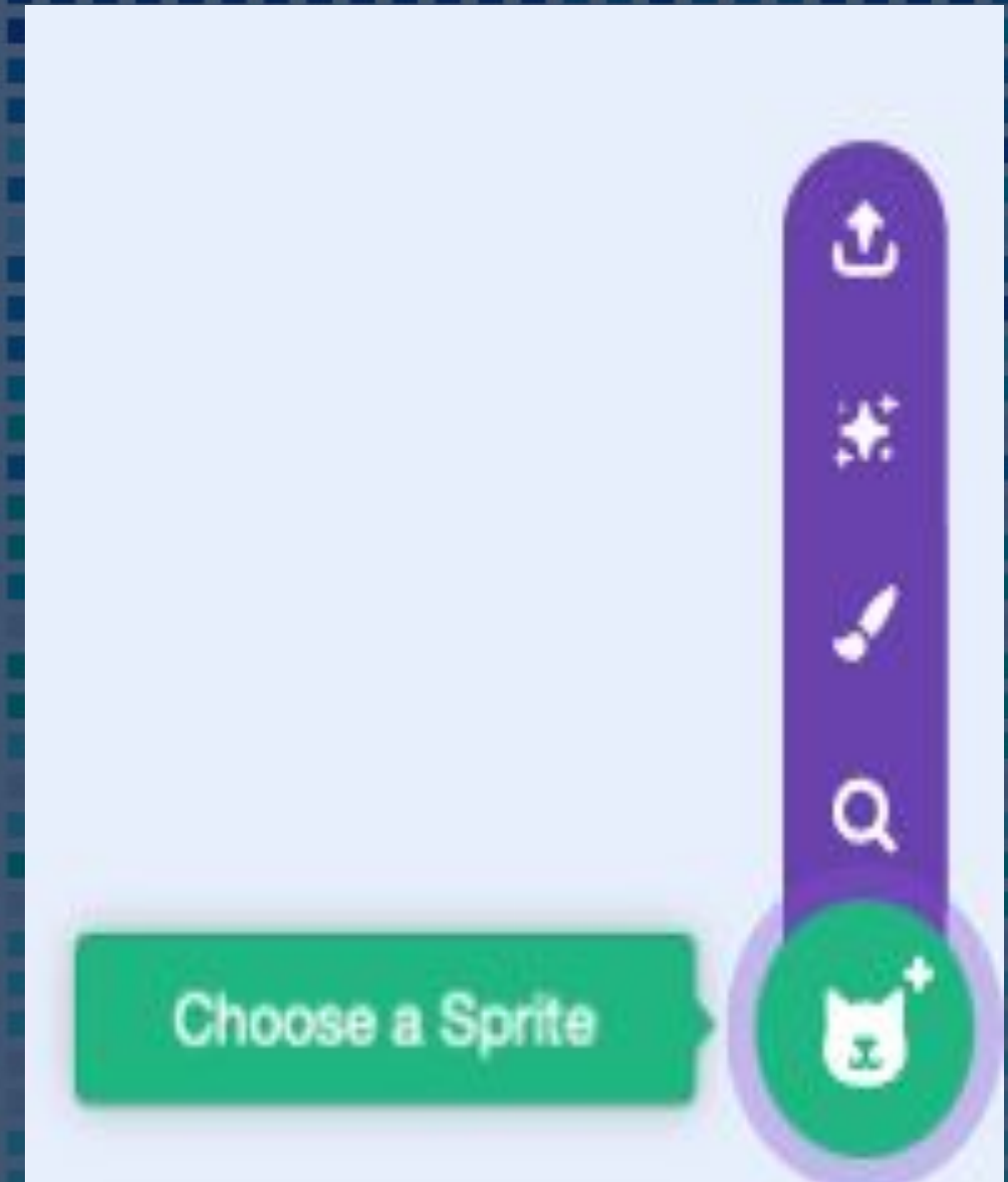
We will look into these more!

What about Scratch Cat?

On this area of our project we have our **sprites**.

Sprites are 2D bitmaps in a large scene... but in simpler terms for us, it is an image that can be programmed to move, respond to events, and interact with other sprites.

We can create sprites in Scratch with the paint brush icon or choose pre made ones! If you make some outside of Scratch you can also upload them!

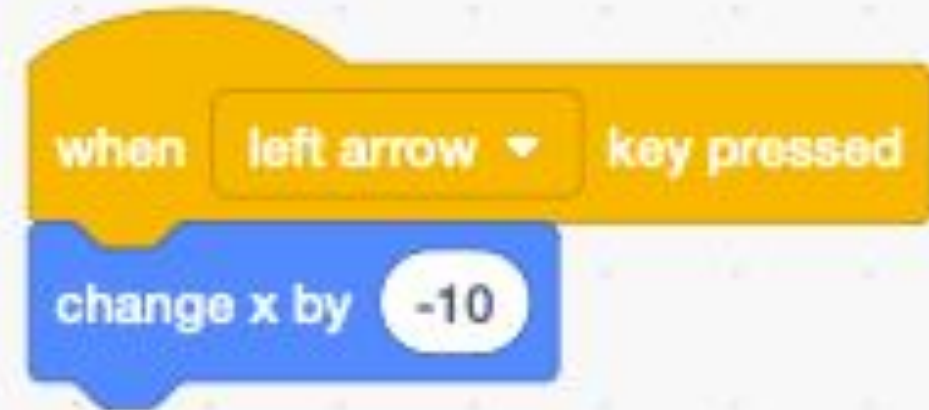
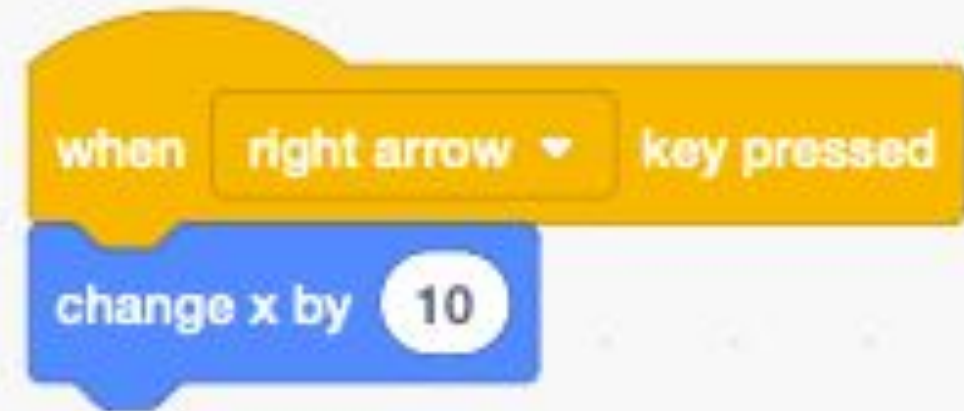


Movement Activity

Goal: Make a sprite move with inputs from the keyboard

Look around the code blocks for something that might make our sprites move.

What do we think that these do?
Try implementing them and then running them on your own!
Try changing the numbers around too! What happens?



How do we count the steps?

Now that we have our sprites moving, let's take it one step further. The question is how do we count the number of steps our sprite has walked.

We are going to introduce a **variable** to help us hold the total number of steps.

A **variable** is named value or data storage that can hold a value or information that can be changed when you execute a program.

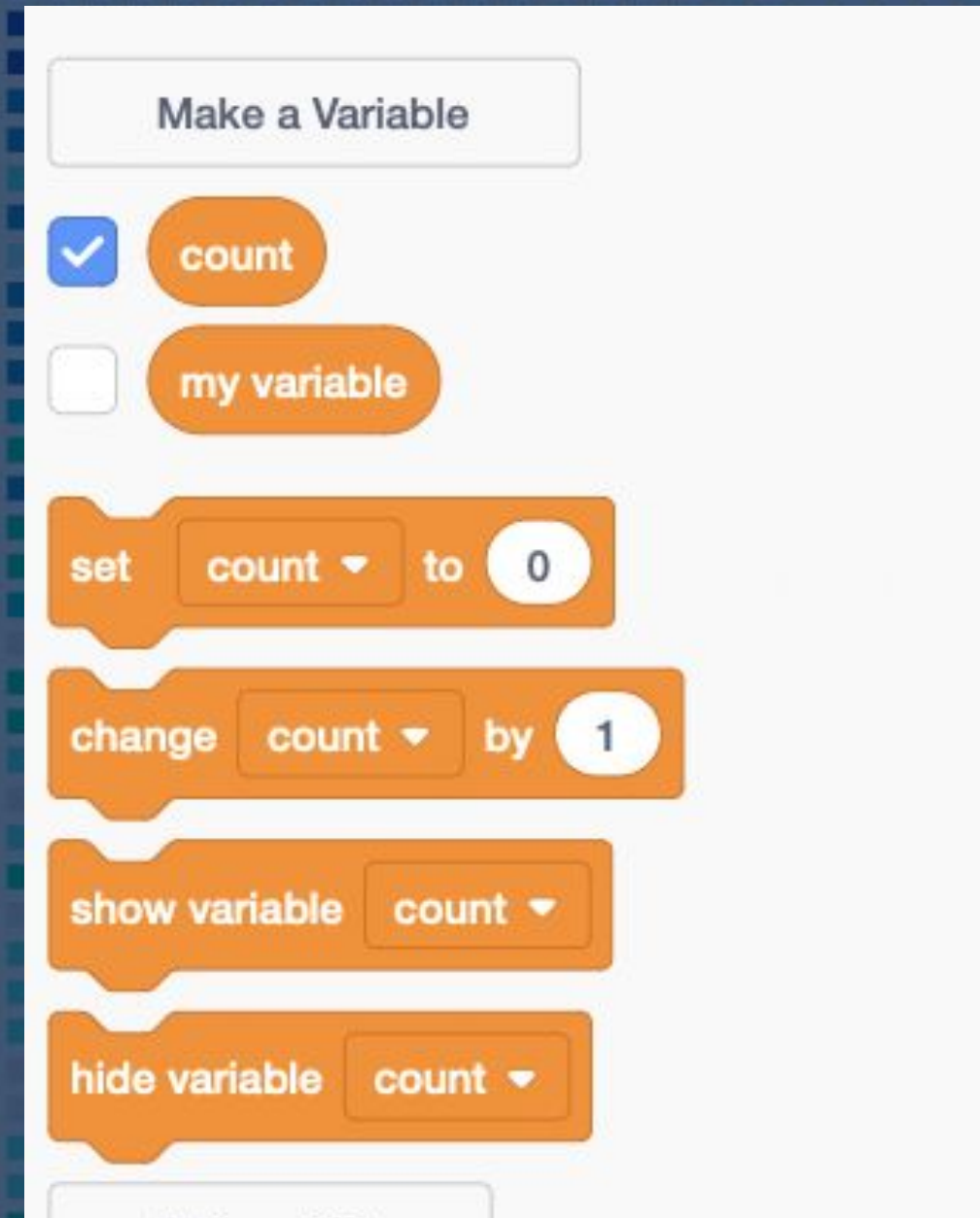
We can use an analogy to help us understand:

Variables are like boxes as we can store many things in boxes. We can label boxes thus telling you what is inside them. Like we can have a box for "Books" in your room where you keep books. You can take books out and add more to the book box. You can have other boxes with toys or clothes too.

Variables in Scratch

We can create variables in Scratch, by going to the orange Variables section in the code blocks.

Notice some of the options we have with our variables, we can set variable to value, change the value, show or hide the variable as well.



Variable Activity

Goal: Make a variable that holds a number

Lets upgrade your program to count number of times you hit an arrow button.

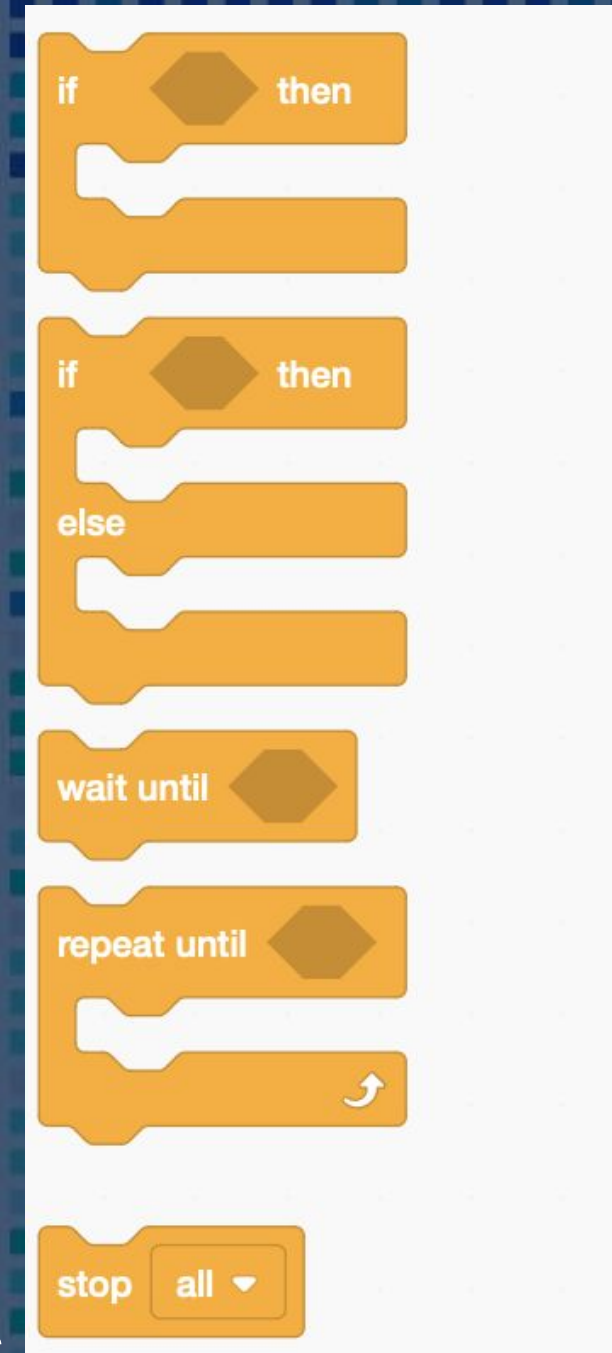
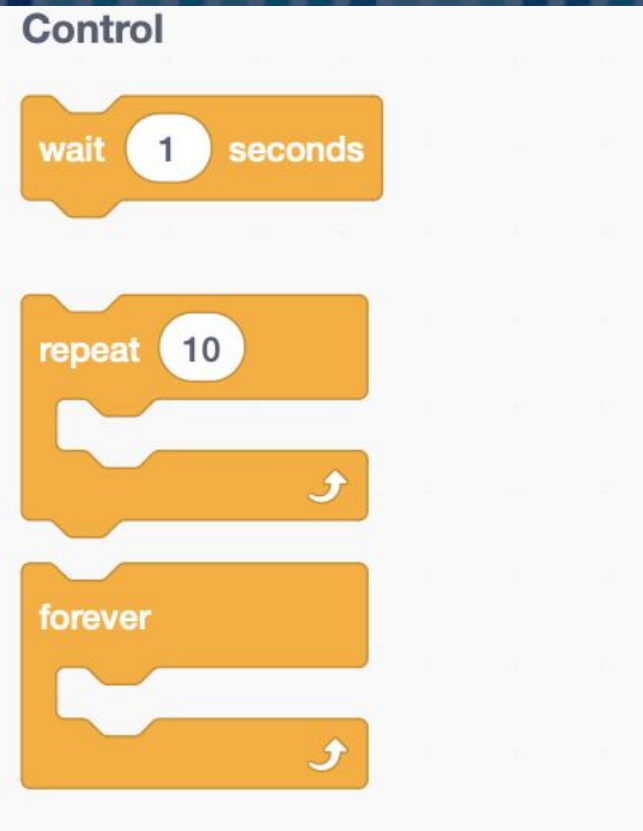
Try getting the count on one direction, then maybe the others
Try using some of the other code blocks! Notice what happens?



Control Structures

Control structures/statements enable a programmer to determine the order in which a program statements are executed. They usually allow for two things: 1. skip some statements while executing others 2. repeat one or more statements while some condition is true

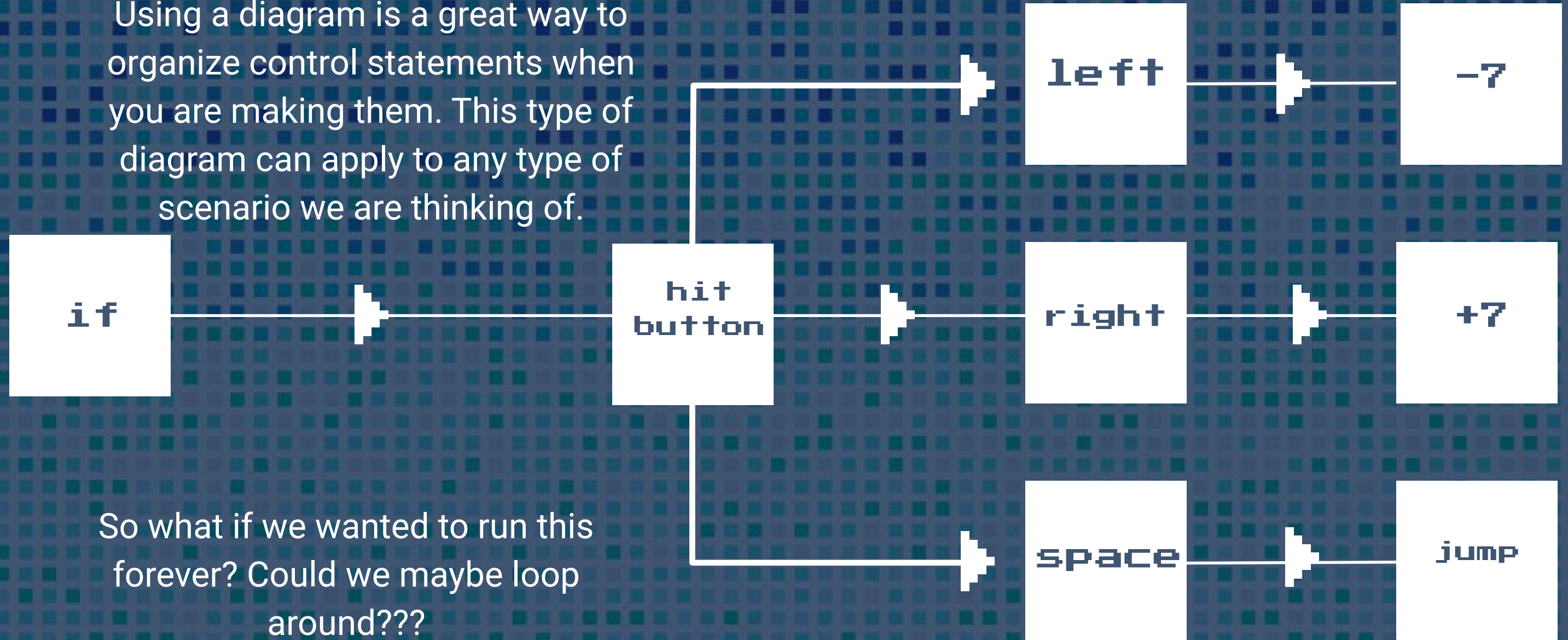
Control structures are all within the Control code block section in our side panel. These come in forms of if's, loops(such as forever, wait, wait until, repeat until) and so on. They work hand in hand with our **sensing** code blocks but we will focus on a few to start.



Let's break it down

but looking at our player movement

Using a diagram is a great way to organize control statements when you are making them. This type of diagram can apply to any type of scenario we are thinking of.



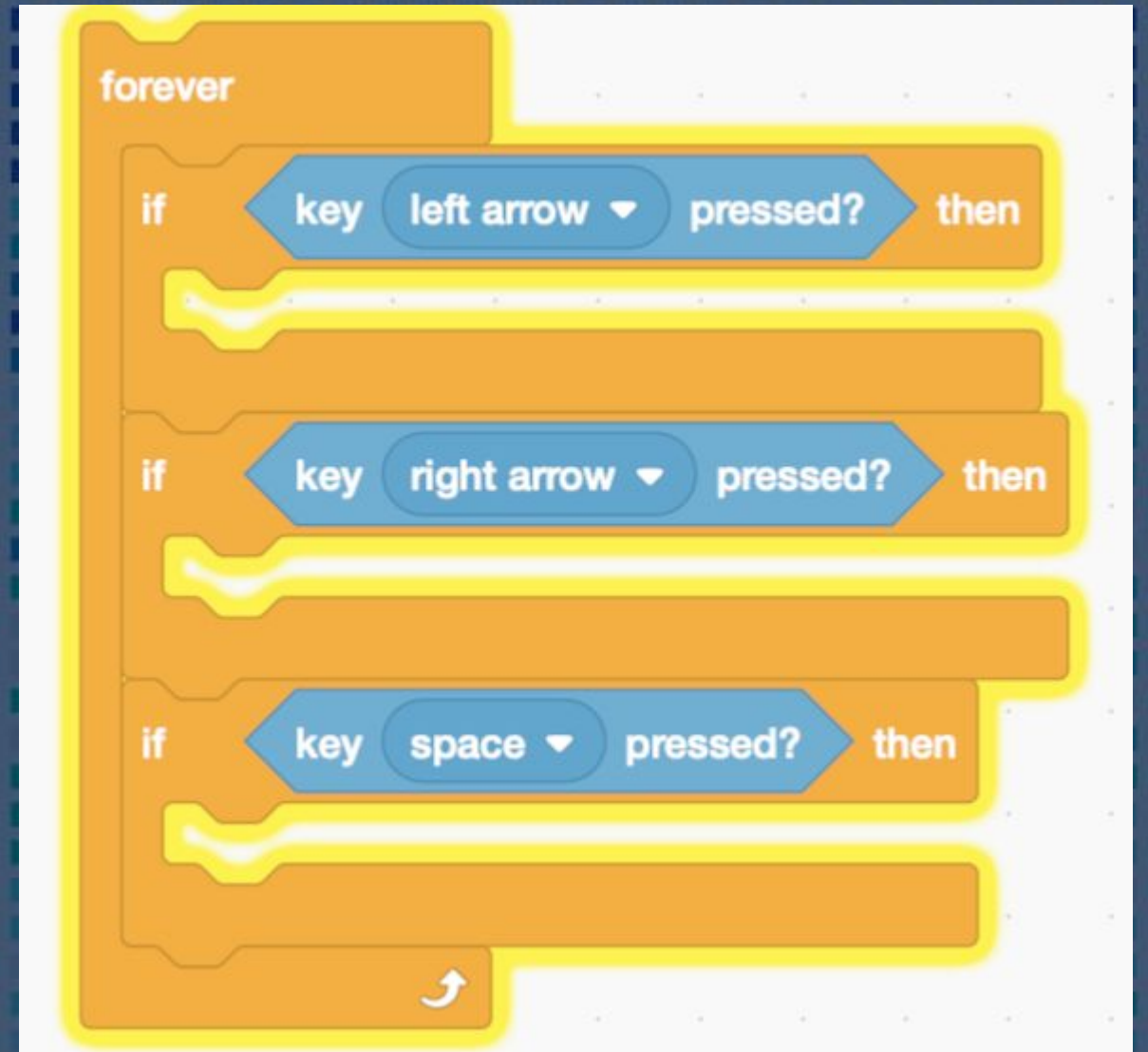
So what if we wanted to run this forever? Could we maybe loop around???

Control/Loops Activity

Goal: Change our code blocks to have control statements

Lets upgrade your program one more time for efficiency purposes by making our movement repetitive.

Try making your code loop or repeat, if you were to hit the green flag! How do you think we would do this? What code blocks could we use



Hint: this is what the diagram looks like but with code and a loop.

Example code

What the code could and might look like remember that all code is different! For example this one's sprites move as well!

